

A Preliminary NASA Compliant Conformal Coating for Optimized Space Radiation Shielding Configurations and its Mass Attenuation Coefficients

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INTRODUCTION

As the next generation of nuclear reactors continues to advance, and with the expanding nuclear and radiation industries in areas such as space operations, waste management, medicine, and imagery, there is an increasing demand for optimized radiation shields. This bulk radiation shielding material must be optimized by minimizing dose to humans, environment, or sensitive electronic equipment, while also designing for cost, volume, and material preparations (Cao et al., 2020).

As polymer applications and popularity grow in various contemporary industries, they are also being developed as matrices for high atomic number (Z) fillers in the field of radiation protection (Long et al., 2022, Kacal et al., 2020). While the traditional and most effective radiation shields remain dense and high Z materials such as lead, or large deposits of relatively cheap materials such as concrete, composites consisting of polymer with high Z fillers compete with or outperform traditional shielding materials in areas such as cost, compatibility, and toxicity (Chang et al., 2015). The addition of metal oxides such as tungsten oxide (WO_3), or gadolinium oxide (Gd_2O_3) powder significantly increase photon shielding worth of polymer-based materials, particularly in the energy range of X-rays (DeVanzo et al., 2020). This is primarily due to the increased likelihood of Photoelectric interactions in materials with higher atomic numbers. As photon energy increases, and Compton scatter becomes the dominant physical interaction between photons and matter, the shielding benefit is reduced.

Polymers are already in use across a wide range of industries. By infusing high Z metal oxide powders during the polymer's preparation stage, the polymer composite can serve as an effective radiation shield while retaining its original design purposes. (Hanson et al., 2021a). The polymer still serves as a flexible, cheap, lightweight, and easily processed matrix, with which the metal oxide powder is infused via one of numerous synthesis methods, such as in situ synthesis (More et al., 2021). In the case of sensitive electronics, the polymer-based conformal coating lines the board's exterior and protects against corrosion, provides insulation, and reduces mechanical stress to its interior circuit board. On integrated circuit boards in spacecraft, it also prevents the growth of whiskers. With high Z filler introduced into the conformal coating, it can additionally

serve in a new purpose of shielding against high energy photons. In some cases, metal oxide infused conformal coating (MOICC) may enhance radiation protection versus the same mass of aluminum, a traditional space shielding material, for Van Allan Belt electrons and protons.

This report demonstrates significant progress towards a prototype homogeneous radiation shielding version of standard conformal coating for spacecraft electronics that is optimized to improve radiation shielding for less sensitive electronics, while closely maintaining the requirements of current industry conformal coating standards (NASA, 2016).

Before a specific MOICC can be a feasible radiation shield, it must first have defined and validated radiation shielding properties. This allows expected performance levels to be incorporated into material selection and design. The primary metric of photon shielding worth is mass attenuation coefficient (μ_m or $\frac{\mu}{\rho}$). This report provides mass attenuation coefficients for two attractive MOICC options for energies ranging from 123 keV to 1.2 MeV, based on high Z oxides and availability with low cost.

MATERIALS AND METHODS

This experiment develops the mass attenuation coefficients (μ_m) of two metal oxide infused polyurethanes (PUR) configured to provide the parallel beam good geometry in which the Lambert-Beer law (Eq. 1) is applicable (Shultis et al. 2000).

$$\mu_m = -\frac{\ln(\frac{I}{I_0})}{x \cdot \rho} \quad (1)$$

Here, μ_m is in units of ($\text{cm}^2 \text{g}^{-1}$). I represents the intensity of attenuated photons, and I_0 corresponds to the intensity of unattenuated photons. The product, $x \cdot \rho$ is aerial density (in g cm^{-2}), x is thickness (cm), and ρ is density (g cm^{-3}). For standard error propagation of the Lambert Beer Law, the uncertainty may be accounted for in Eq. 2. On notation, $\sigma(x)$ represents the standard deviation (σ) of the given variable, x .

While there are slight alternates in the reporting of error propagation of the Lambert-Beer Law (Singh Mann et al., 2016, Akman et al., 2018), Eq. 2 was derived from the Lambert-Beer law using propagation of error techniques provided by (NIST, 1966):

$$\sigma(\mu_m) = \frac{1}{\mu_m} \sqrt{\left(\frac{\sigma(I_0)}{I_0}\right)^2 + \left(\ln \frac{I}{I_0}\right)^2 \times \left(\frac{\sigma(\rho x)}{\rho x}\right)^2}$$

(2)

The Lambert Beer law is only applicable in the case of a “good geometry” or “narrow beam” configuration (Shultis et al. 2017, Al-Buriahi et al. 2019). Therefore, it is necessary to ensure a parallel pathway for photons to pass through a known thickness of material before entering the face of a detector.

Due to the isotropic nature of radioactive decay, there is a nonzero presence of slightly unparallel photons leaving a source and entering a detector; when good geometry is achieved, this is a negligible amount. When possible, collimators consisting of dense, high Z materials should surround the experimental setup to minimize detection of secondary radiations and background. There is even greater necessity for High Z collimators in cases of poor detector energy resolution. Furthermore, good use of collimators improves detector energy resolution at particularly low ionizing photon energies (e.g., 10-30 keV) such as the lower probability photons generated in the decay of ^{241}Am (Akman et al., 2018).

The Micro-Detective High Purity Germanium (HPGe) Detector was utilized for this experiment for the exceptional energy resolution that it provides. The source and shield are placed as close to the detector as possible to maximize detection counts over the shortest possible time, while also accounting for narrow beam geometry considerations.

A Europium (^{154}Eu) source was selected due to its wide range of decay photon energies from the traditional X-ray energy regime into the MeV gamma energy regime. This makes it a useful comparison for Bremsstrahlung X-rays from slowing charged particles, for artificial fission environments such as reactors, and for shielding high energy gamma sources. For the case of space radiation shielding, this composite material is optimized to compete with Aluminum of comparable mass for Van Allen Belt electrons and protons and to outperform Aluminum in X-ray photon shielding. There are currently no feasible shielding technologies for the high energy galactic cosmic rays, and damage from these radiation sources are mitigated against via other means.

The base polymer composite explored in this report is ReoFlex 50 Smooth-On, a popular and inexpensive two-part resin and hardener polyurethane material. It was filled with an aggregate of metal oxide powder. As a benchmark and for comparison, pure aluminum slabs were also measured for mass attenuation values in these experiments. In this experiment, the specific MOICC shields under consideration were that of pure ReoFlex 50, 50 wt.% tungsten oxide infused into ReoFlex 50, and 60 wt.% gadolinium oxide infused into ReoFlex 50. The MOICC materials were mixed directly via in situ synthesis and poured into molds for curing.

The mixing ratio of Part A (resin) to Part B (hardener) is 1:1 by volume. For the metal oxides, Gd_2O_3 particles

with $< 10 \mu\text{m}$ in diameter (Beantown Chemical Corporation) and WO_3 particles with $10 \mu\text{m} - 20 \mu\text{m}$ in diameter (Beantown Chemical Corporation) were selected. The metal oxide particles were first added to Part B and the mixture was stirred until observable uniformity. Part A was then added to the mixture, followed by stirring for a uniform mixture. The mixture was then poured into a mold for the desired shape. Vacuum degassing was then performed to reduce the air bubbles trapped inside the mixture. At last, the mixture was cured at room temperature for 12 hours for final MOICC samples.

This preparation method has considerable improvements over previous generations (Hanson et al., 2021b). As observed under a microscope in Fig. 1, these shielding materials feature considerable homogeneity when qualitatively compared to identical scale images of previous generations (Hanson et al., 2021b). In addition, the MOICC samples have minimal cavities due to the vacuum degassing process, which has led to visible decrease in cavities observed as compared to previous generations. White gadolinium oxide, in which particle mean diameter is approximately $5\text{-}10 \mu\text{m}$ according to the manufacturer, showed indistinguishable homogeneity at this scale. Even the larger particle sized tungsten oxide with mean particle diameter of approximately $60 \mu\text{m}$ according to the manufacturer, supports the expectation of a largely uniform distribution inside of the PUR.

Generally, finer MO particle size and increased particle purity will coincide with higher MO price. It is important to ensure material homogeneity and to minimize the presence of air bubbles in the material for both radiation shielding effectiveness and other material properties (Li et al., 2017). Sample images of the shielding pieces are shown in Fig. 2 and Fig. 3.

Theoretical values of composite material mass attenuation coefficients, $\frac{\mu}{\rho_{\text{composite}}}$ may also be determined via the weighted sum of various element's mass attenuations as in Eq. 3, in which the weight percent w_i is multiplied by the mass attenuation of the i^{th} constituent element $(\frac{\mu}{\rho})_i$ in the compound.

$$\frac{\mu}{\rho_{\text{composite}}} = \sum_i w_i (\frac{\mu}{\rho})_i \quad (3)$$

In theoretical calculations, a generic polyurethane formula ($\text{C}_{17}\text{H}_{16}\text{N}_2\text{O}_4$) was utilized. Although this is not the exact chemical composition of ReoFlex 50 (which is kept as a manufacturer trade secret), this will not make a notable difference in attenuation modeling because of the nearly identical photon shielding performances of the similar low Z elements in a polymer mix (NIST, 2004).

Theoretical calculations do not account for potential heterogeneity in materials, in case of experimental shielding underperformance versus theoretical expectations, a potential source of this underperformance is filler gaps in the polymer matrix leading to considerable heterogeneity.

With the reporting of mass attenuation values, one can quickly determine other equivalent metrics for a material's shielding properties at the given photon energies, such as half or tenth value thickness, or mean free path. In further photon shielding modeling, design, and potential deployments, photon buildup considerations are essential in addition to common metrics for $\frac{\mu}{\rho}$.

The open-source software package, Cambio was utilized to process all spectra. In Cambio, automatic gaussians are generated over energy peaks; these gaussians provide estimated counts areas and uncertainties that represent photon intensities (USDOE, 2015). Unattenuated and attenuated photon intensities are determined via photo-peak count areas for live-time equivalent exposures without and with shield present, respectively. Fig. 4 features a diagram of the experimental setup.

Thicknesses were determined before radiation with repeated caliper measurements on the shield samples. After the shielding measurements, the density and its associated uncertainty were again determined by cutting shields, then performing weight and water displacement volume measurements on the pieces.

RESULTS AND DISCUSSION

The mass attenuation coefficients with their associated uncertainties are presented in Table I. Note that theoretical energy values in Table II differ from energy values in the experiment due to the decay specifics of artificial Europium. These values are also visualized in Figs. 5-8.

The Aluminum benchmark values trend with NIST reported attenuation coefficients are shown in Fig. 5. This provides experimental setup validation for the measured values for our new material, MOICC. Furthermore, as observed in Fig. 6 and Fig. 7, MOICC measured values coincide with theoretical predictions. This benchmark allows for reasonable predictions of other potential combinations of metal oxide and polyurethane.

The metal oxide infused polyurethanes both show an order of magnitude mass attenuation improvement over pure polyurethane and Aluminum at X-ray energies, as observed in Fig. 8. This figure represents comparisons between all experimentally measured PUR and PUR composites. The addition of either metal oxide as filler inside a device's external conformal coating will significantly reduce X-ray dose to internal sensitive electronic devices.

Experimental values at various energies are interpolated using the summation of two exponential decays that replicate the expected interaction likelihoods of Photoelectric effect and Compton scattering, the two dominant physical interactions between photons and matter in the energy regime of the Europium source (0.1 to 1.2 MeV). This method of curve fitting, which utilizes the Levenberg-Marquardt algorithm (Hayes et al. 1998), results in a closer match to the known physics than the more

commonly observed straight-line connection between points.

The addition of either of the two metal oxide fillers is shown to increase the photon shielding worth of a coating material. However, tungsten oxide of 50 wt.% or higher is the recommended choice for optimal X-ray shielding. The tungsten oxide shows improved photon shielding worth over the other examined metal oxide. At energies above ~500 keV, and when limited to small amounts of material such as that of spacecraft applications. As such, high shielding effectiveness remains unattainable here, and successful shielding in spacecraft at these high photon energies beyond 500 keV would require an associated mass penalty.

The benchmarking of experimental mass attenuation values of MOICC with theoretical predictions confirms that the current production method of MOICC samples produces sufficient levels of homogeneity for use in shielding design. Thus, further theoretical calculations may allow for shielding optimization with other potential metal oxide fillers as long as materials quality does not change, and homogeneity remains exceptionally high.

CONCLUSION

While Tungsten oxide proves more efficient shielding photon worth at a lighter wt.% than gadolinium oxide, this shielding improvement may not be substantial enough to eliminate gadolinium oxide based MOICC in future applications. This is because the gamma shielding improvement could be negated by neutron shielding considerations in reactor shielding or other artificial source applications. Gd may be particularly attractive in neutron shielding applications due to its enhanced thermal neutron cross section (Long et al., 2022). Similarly, in natural environments, other types of radiation shielding worth will differ slightly from these outcomes, such as with various electron spectra.

The addition of tungsten oxide filler shows an effective way to improve X-ray shielding worth by multiple factors at the 123-247 keV energy range versus pure ReoFlex coating or versus Aluminum. Similarly, gadolinium oxide infusion also effectively improves X-ray shielding worth in this energy regime. These MOICC samples featured exceptional homogeneity that allowed for performance levels to match theoretical calculations. With the low cost of the metal oxide powder and polyurethane coating, these materials show promise as cost-effective radiation shielding for various applications.

Further research must be done to determine and verify shielding performance against various other types of radiation, and in order to guarantee their other material properties will configure with any proposed application. Following this assessment, follow up studies in both radiation transport codes and in electron, proton, or high energy neutron beams should be performed in order to better quantify shielding worth for outer space applications. These

types of experiments would allow one to predict shielding worth for most types of satellite applications, where the primary radiative particles of concern are electrons from ambient energies up until 7 MeV and protons of a wide range of energies from ambience up until 10s of MeV. Similarly for artificial shielding applications such as reactors, fission neutron shielding worth experiments may be performed in order to compliment gamma shielding assessment for overall effectiveness against artificial environments.

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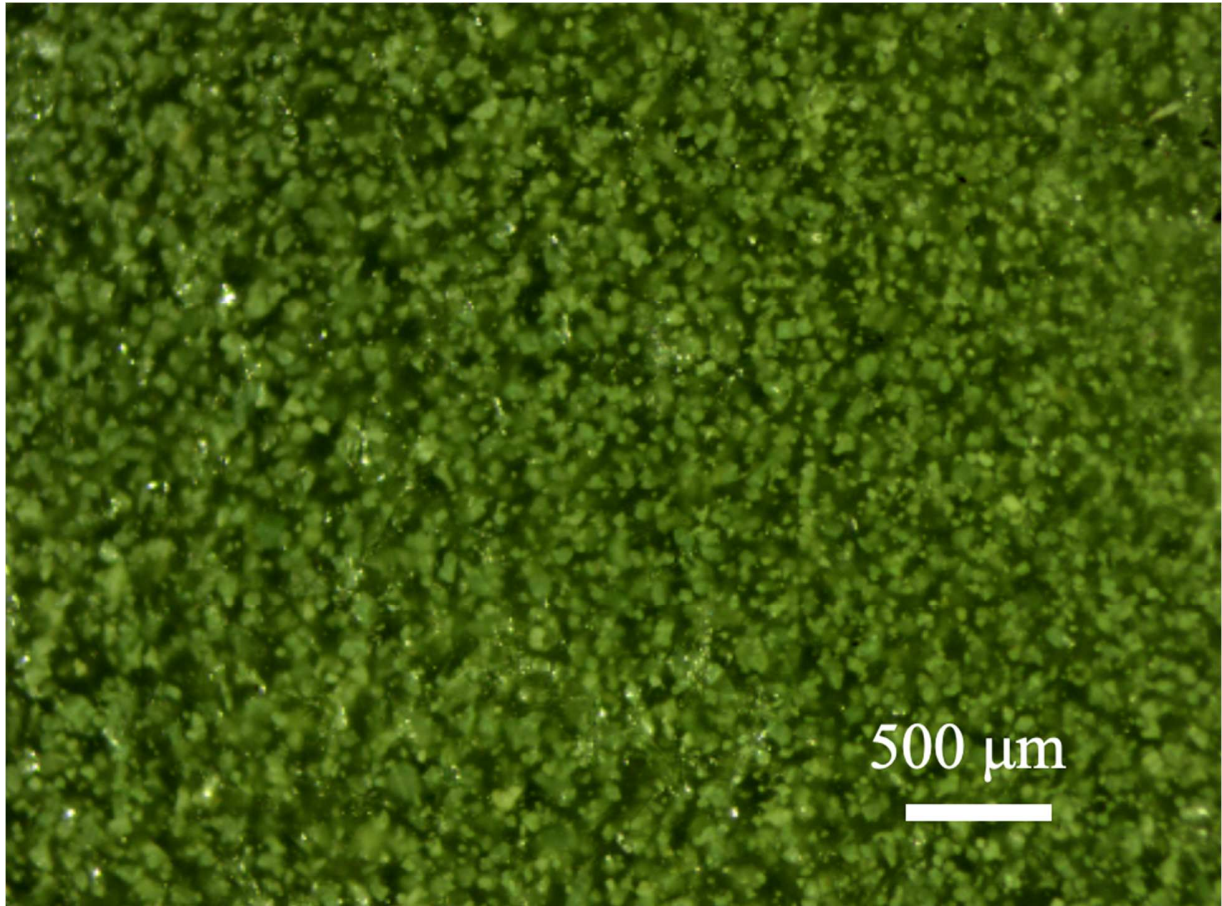


Fig. 1. The tungsten oxide samples feature substantially improved homogeneity over previous generations of MOICC (Hanson et al. 2021b).

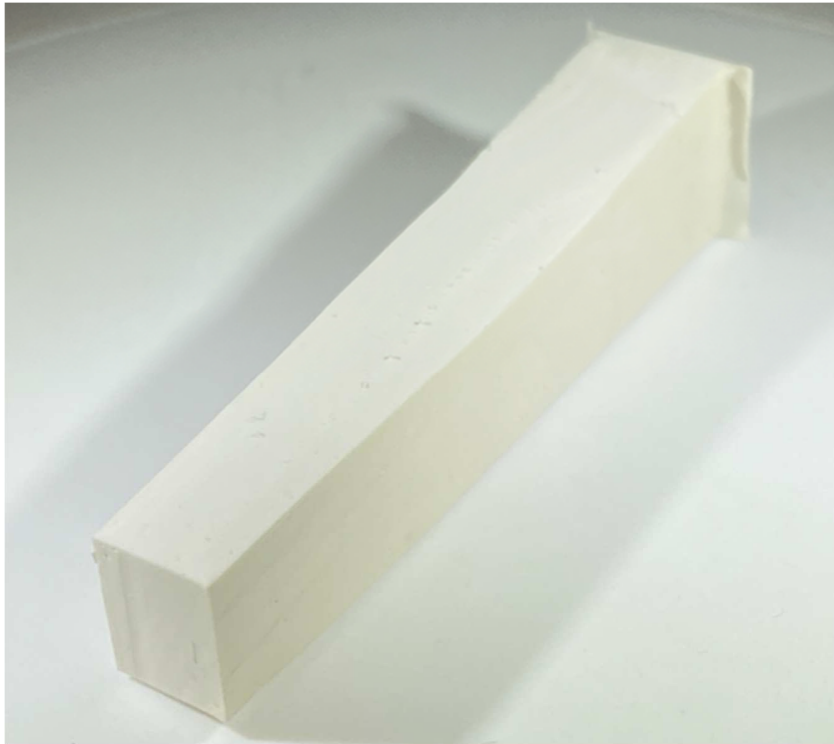


Fig. 2. Macroscopic view of Gadolinium Oxide infused polyurethane sample.



Fig. 3. Macroscopic view of Tungsten Oxide infused polyurethane sample.

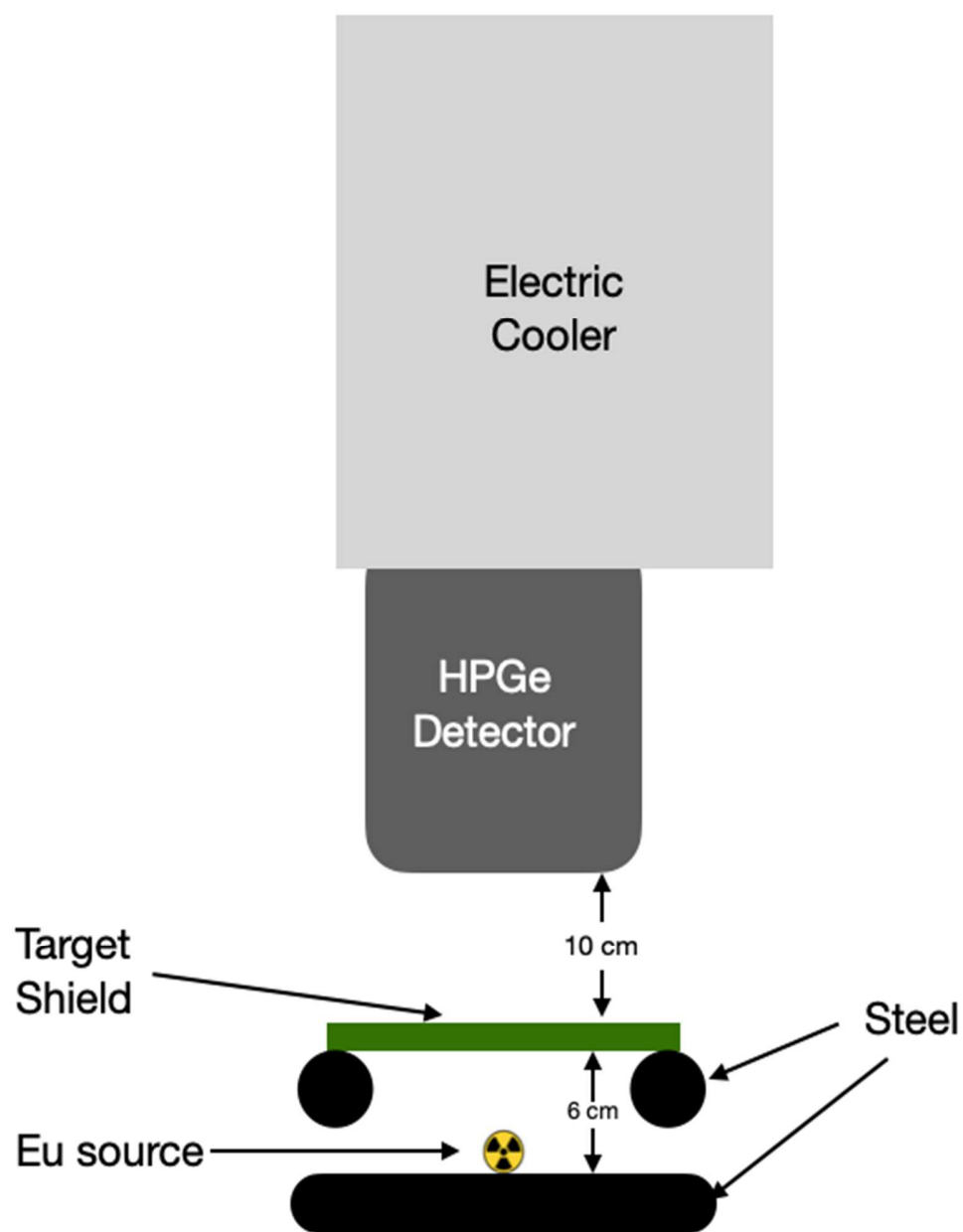


Fig. 4. Diagram of experimental setup. Note the detector is suspended above configuration on a tripod. Figure is not to scale.

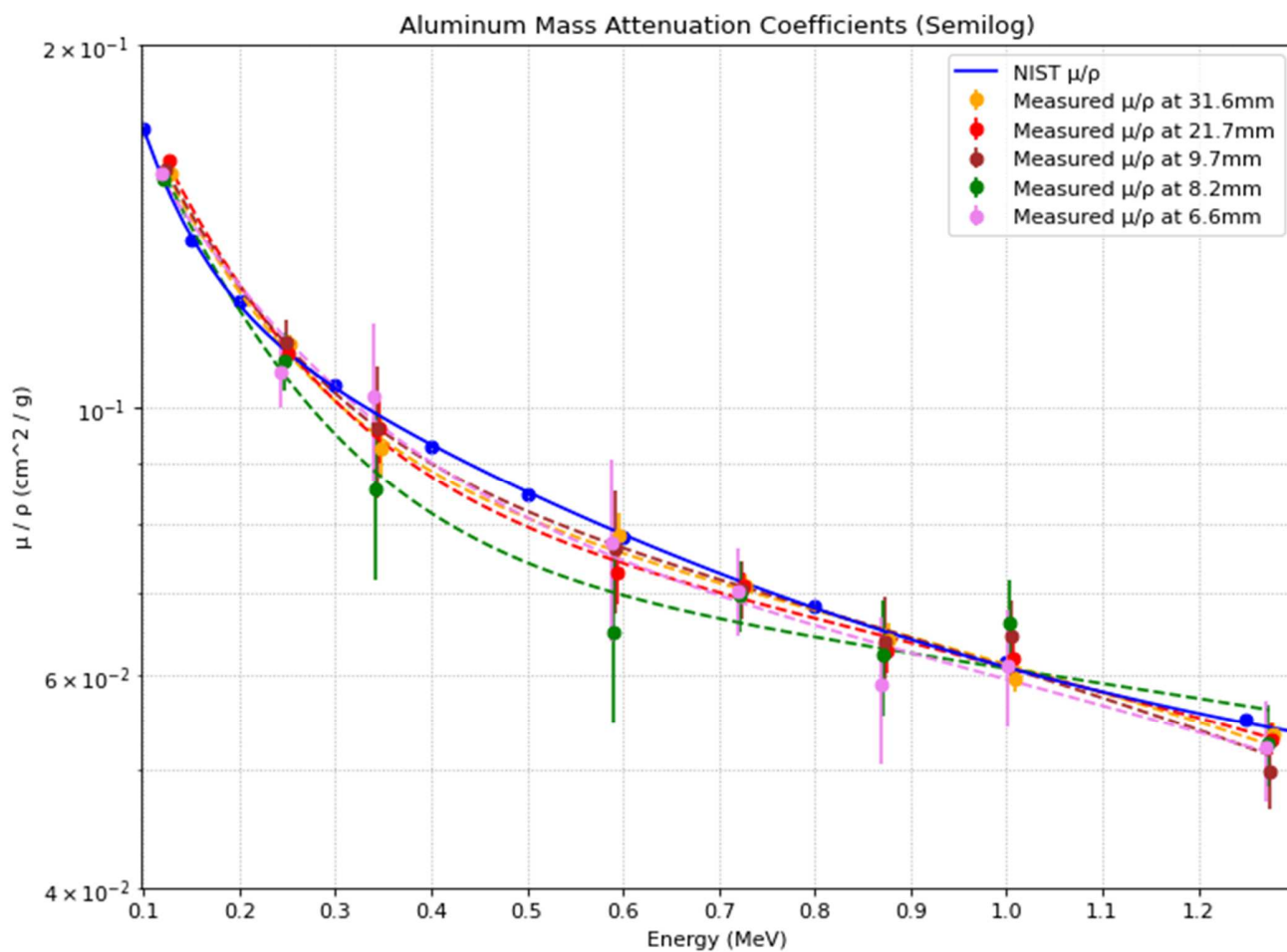


Fig. 5. Measured aluminum mass attenuation coefficients compared to NIST values (log Y axis).

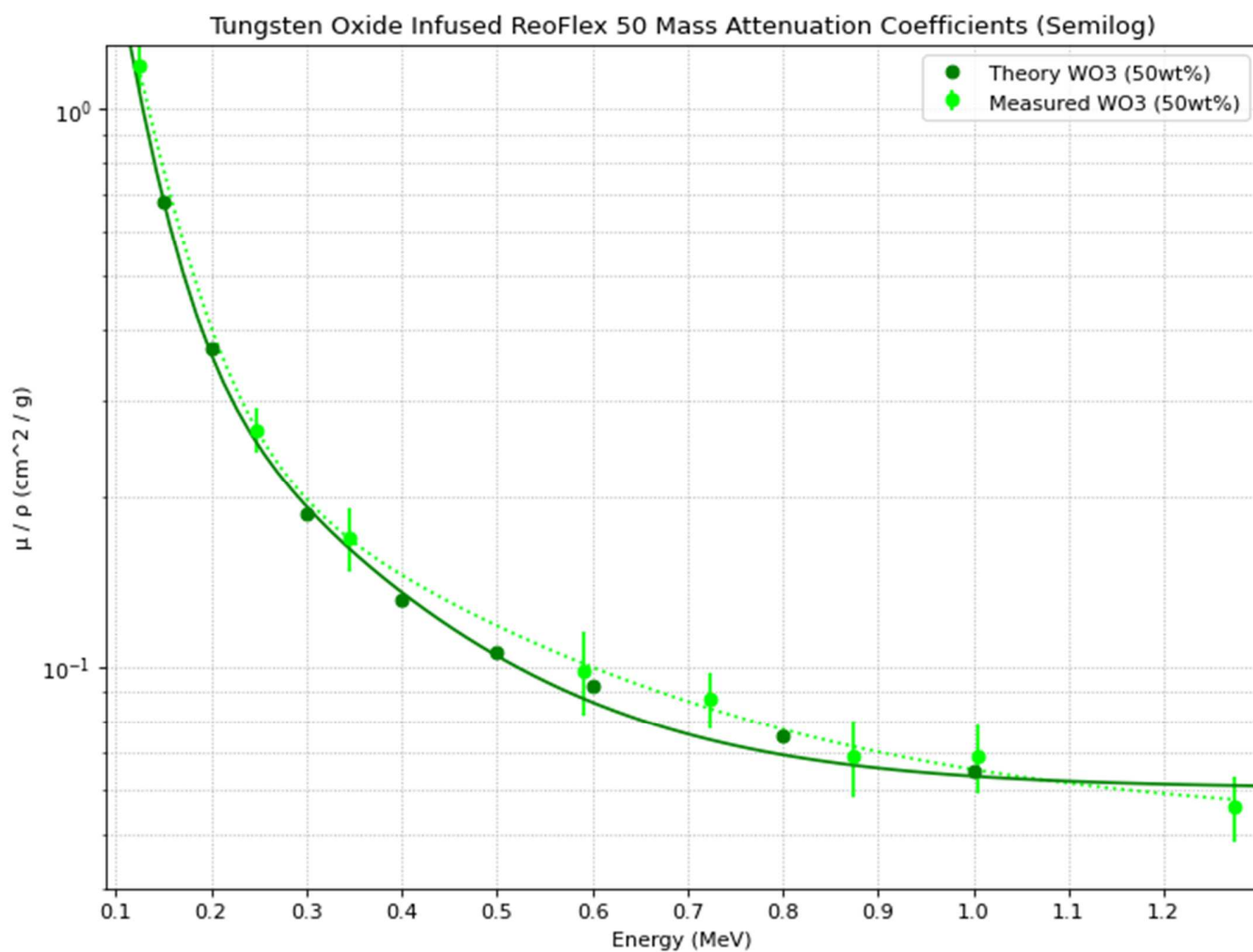


Fig. 6. Tungsten oxide mass attenuation measurements alongside theoretical values (log Y axis).

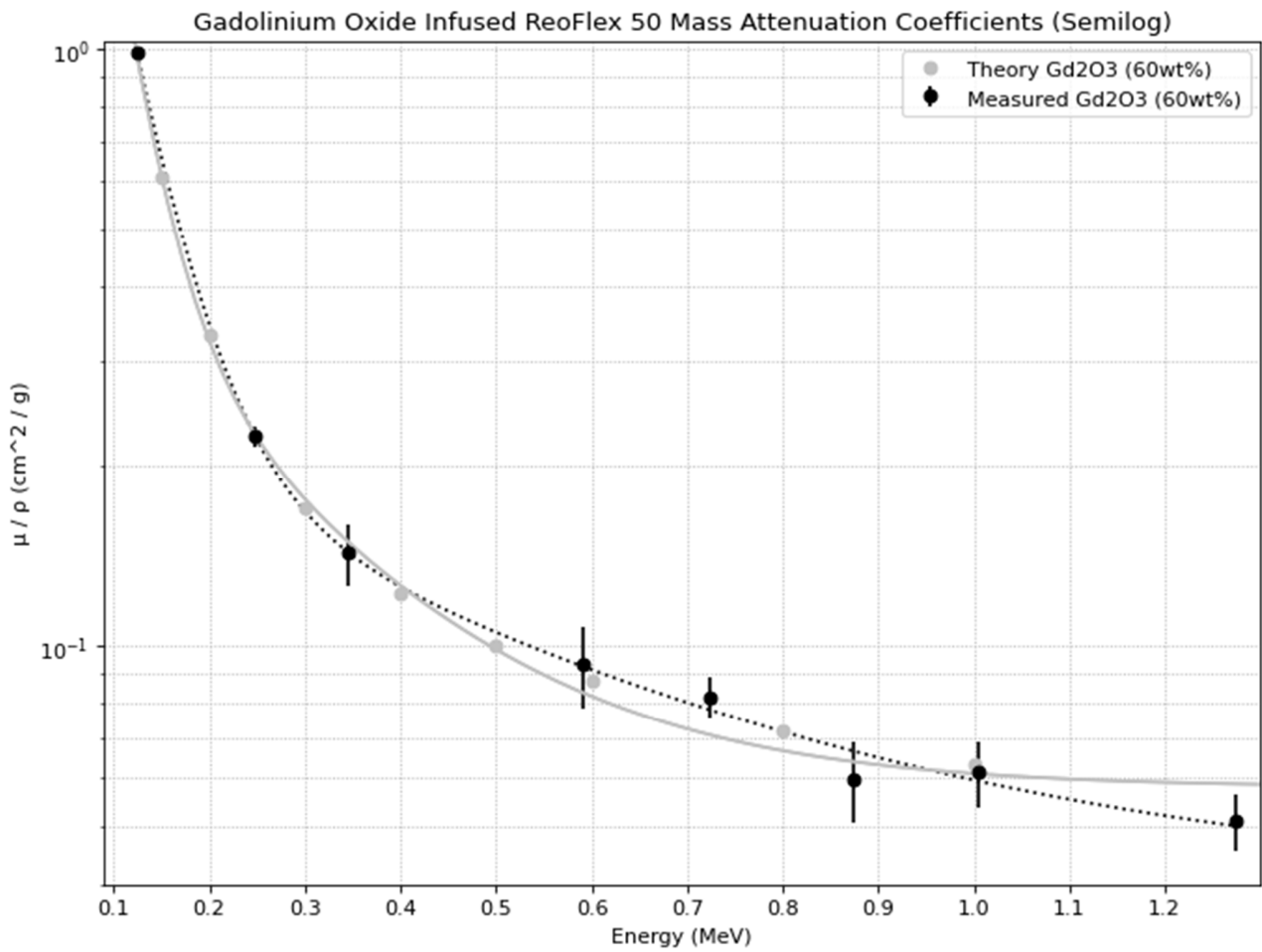


Fig. 7. Gadolinium oxide mass attenuation measurements alongside theoretical values (log Y axis).

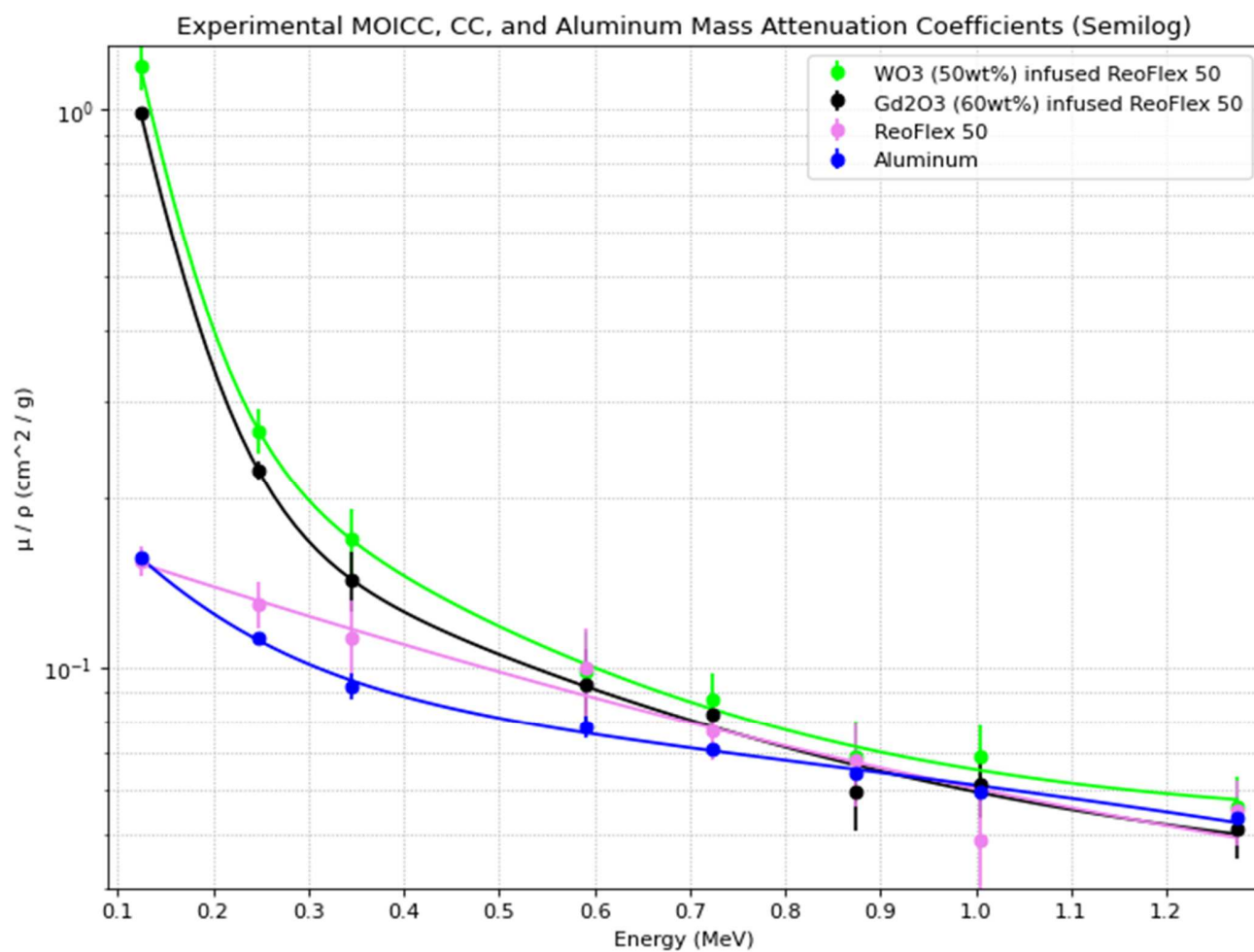


Fig. 8. MOICC mass attenuation measurements (log Y axis).

TABLE I. Mass Attenuation Coefficients (μ_m in $\text{cm}^2 \text{g}^{-1}$) for all material at various energies (MeV)

γ Energy (MeV)	0.1237	0.2479	0.3442	0.5917	0.7233	0.8732	1.0045	1.2744
WO_3 (50 wt%) ReoFlex 50 Composite	1.1881	0.265	0.1699	0.0983	0.0873	0.0688	0.0689	0.0558
Error in WO_3 (50 wt%) ReoFlex 50 Composite	0.1051	0.0247	0.0219	0.0167	0.0099	0.0107	0.0099	0.0073
Gd_2O_3 (60 wt%) ReoFlex 50 Composite	0.987	0.2246	0.1429	0.0928	0.0821	0.0596	0.0612	0.0509
Error in Gd_2O_3 (60 wt%) ReoFlex 50 Composite	0.0112	0.0087	0.017	0.0147	0.0065	0.0091	0.0076	0.0056
ReoFlex 50 Polyurethane	0.1545	0.1295	0.1127	0.0995	0.0765	0.0678	0.0488	0.055
Error in ReoFlex 50 Polyurethane	0.0095	0.012	0.0189	0.018	0.0088	0.0116	0.0098	0.0073

TABLE II. Mass Attenuation Coefficients (μ_m in $\text{cm}^2 \text{g}^{-1}$) for theoretical materials at energies (MeV).

γ Energy (MeV)	0.1	0.15	0.2	0.3	0.4	0.5	0.6	0.8	1	1.5
WO_3 (50 wt%) ReoFlex 50 Composite	1.789	0.68	0.37	0.187	0.131	0.106	0.092	0.075	0.065	0.052
Gd_2O_3 (60 wt%) ReoFlex 50 Composite	1.627	0.608	0.331	0.17	0.122	0.1	0.087	0.072	0.063	0.05
Generic Polyurethane $C_{17}H_{16}N_2O_4$	0.157	0.141	0.129	0.112	0.1	0.091	0.085	0.074	0.067	0.054
Aluminum	0.161	0.134	0.12	0.103	0.092	0.084	0.078	0.068	0.061	0.05